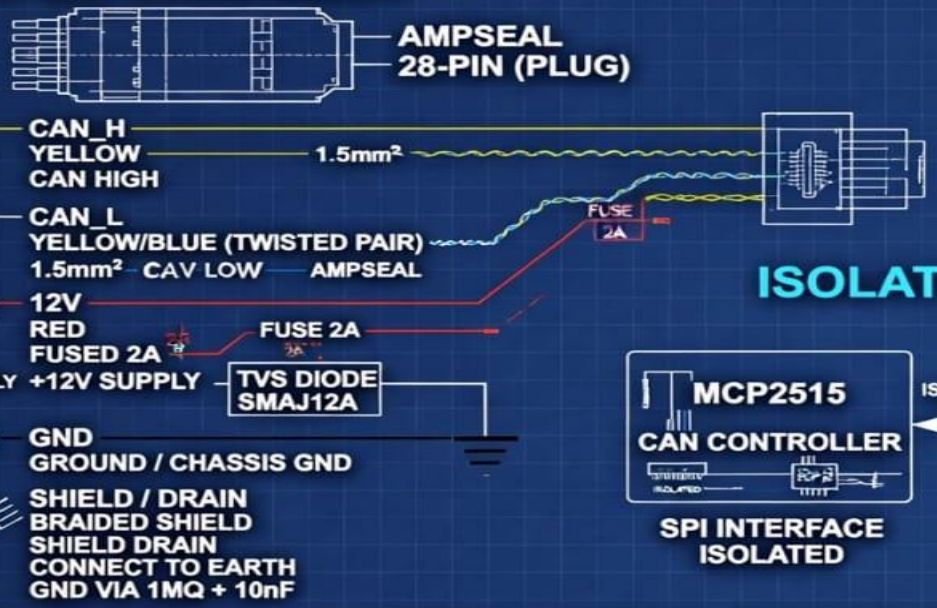
KIA EV3 PESQUERIA - WIRING HARNESS DIAGRAM

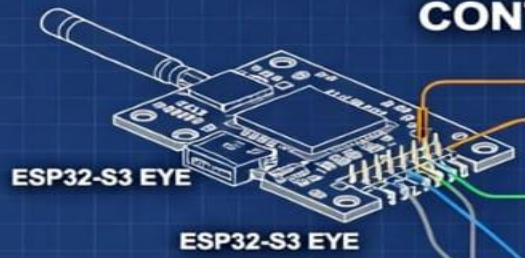
ENGINEER: Norberto Bravo Tadeo, Cheran Michoacan

DRAWING NO: KIA-EV3-PESQ-WH-001 REV B

28-PIN AMPSEAL CONNECTOR — KIA SPEC 18790-02850



VCU ESP32-S3 EYE CONTROL MODULE



SCK — PIN 18 ORANGE 0.5mm² TXL to ESP32-S3 GPIO18

MISO — GREEN 0.5mm² TXL ESP32-S3 GPIO19

MOSI — PIN 23 to ESP32-S3 GPIO23

CS — GRAY to ESP32-S3 GPIO5

ISOLATED CAN BUS INTERFACE



SPI INTERFACE ISOLATED

2.5kV ISOLATION BARRIER

OPTO-ISOLATION INCLUDED

SIGNAL TABLE — WIRING & SPECIFICATIONS

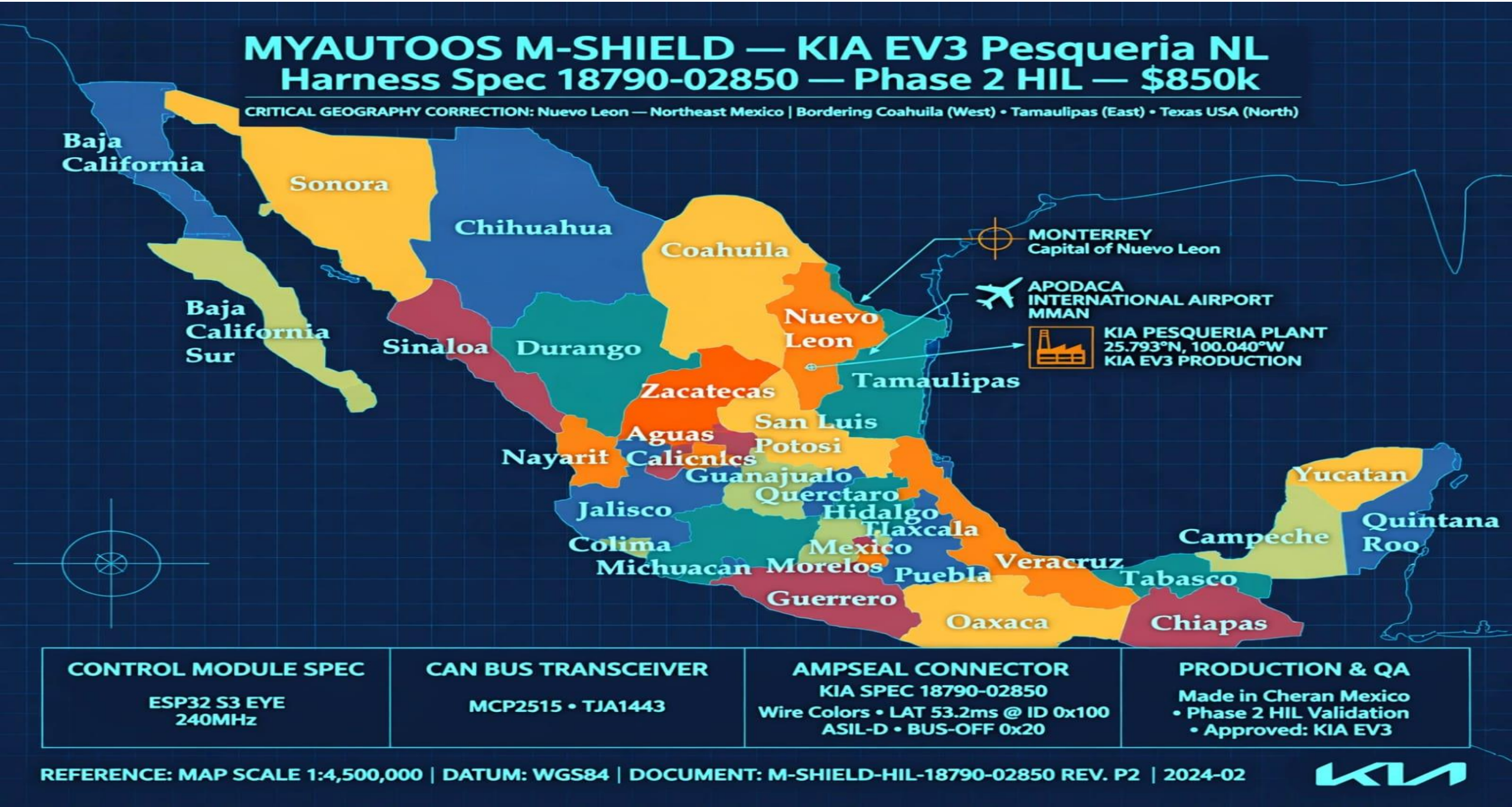
PIN	PIN / SIGAL	WIRE COLOR		GAUGE FUNCTION	DESTINATION / DESTINATION PIN	NOTES
C5	CAN_H	YELLOW	1.5mm²	CAN High Signal	TJA1443 Pin CANH	Twisted Pair to C6
C6	CAN_	YELLOW/BLUE	1.5mm²	CAN Low Signal	TJA1443 Pin CANL	Twisted Pair to C5
A12	12V	RED	1.5mm²	12V Battery Supply	+12V to DC-DC IN	Fused 2A, TVS SMAJ12A
A13	GND	BLACK	1.5mm²	Ground	Fused 2A, TVS SMAJ12A	Chassis Ground
SCK	ORANGE	0.5mm² TXL		SPI Clock	ESP32-S3 GPIO18	Isolated via ISO
MISO	GREEN	0.5mm² TXL		SPI MISO	ESP32-S3 GPIO19	Isolated via ISO
MOSI	BLUE	0.5mm² TXL		SPI MOSI	ESP32-S3 GPIO23	Isolated via ISO
CS	GRAY	0.5mm² TXL	-	SPI Chip Select	ESP32-S3 GPIO5	Isolated via ISO
	SHIELD/DRAIN	BRAIDED	-	Shield Ground	GND via 1MΩ/10nF	Isolated to Chassis

LEGEND:
- CAN_H
- CAN_
- C2V
- GND
- Isolation Barrier
- braided shield

NOTES:
- All twisted pair CAN wiring shall be 120Ω characteristic impedance, twist rate 20-25 twists per meter.
- Isolation barrier rated 2.5kV RMS for 1 minute between low-voltage logic and vehicle bus.
- TVS diode SMAJ12A provides transient protection on 12V line, clamp 19.9V.
- Shield drain to be grounded at VCU end only to prevent ground loops.
- Connector sealing: IP67 rated, use silicone seal on AMPSEAL.

REVISION HISTORY
REV B — 2024-01 — Initial Release — N.B. Tadeo

2 - Mapa Pesquería:02_KIA-EV3_Pesqueria_NL_Map_Harness_Spec_18790-02850_Phase2_HIL.pdf



3 - Wokwi Sim:03_MyAutoOS_Wokwi_ESP32_Simulation_GPIO2_Blue_GPIO4_RED_SAFE-CUT.pdf

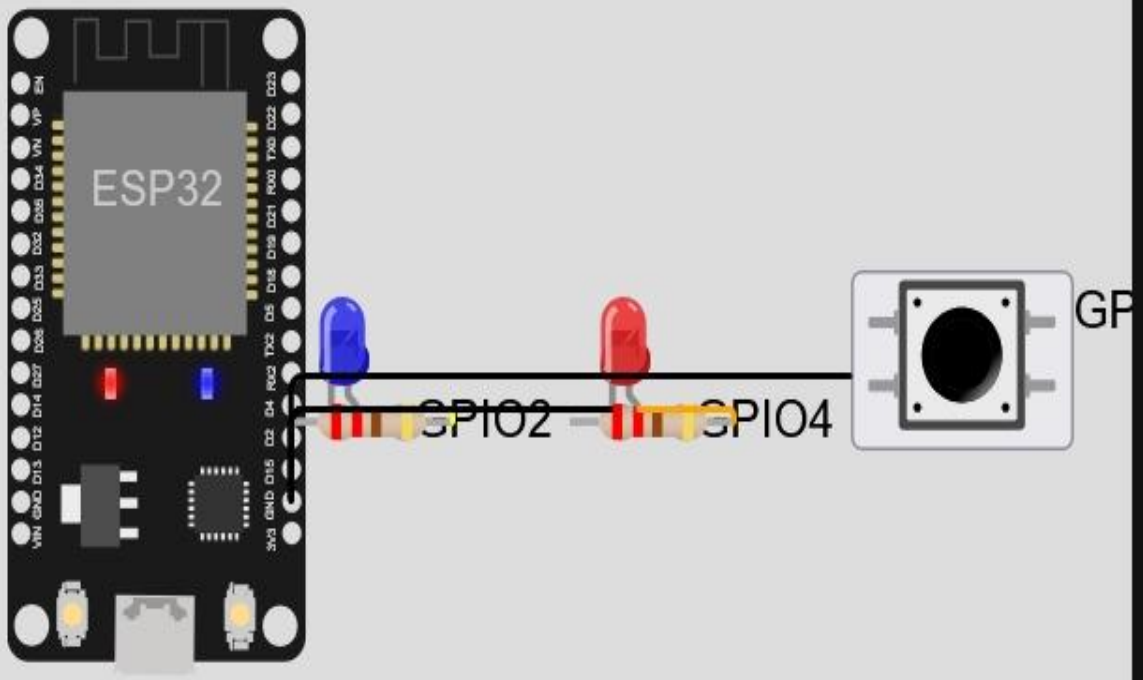
WOKWI SAVE SHARE Docs SIGN UP

sketch.ino ● diagram.json ● Library Manager

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Simulation

02:06.296 64%



MyAutoOS
TECHNOLOGIES

PROVEEDOR
TIER2 • AUTOMOTIVE

MyAutoOS Technologies - Proveedor Tier2
KIA Pesqueria EV3 2026

Founder: Norberto — Cheran, Michoacan

COST ADVANTAGE

BOM
$645 MXN PoC

↓ vs Bosch **$42,500 MXN**

65x cheaper

Cost reduction validated on PoC hardware

REAL-TIME PERFORMANCE & SAFETY

LAT 53.2ms • ASIL-D <60ms KIA Spec
Latency compliant to KIA requirements

BUS-OFF Real • EFLG_TXBO 0x20

SAFE-CUT GPIO4 • Fault isolation validated

PROGRAM STATUS

20 VCUs
HiL Contract
$850k MXN • 2 weeks delivery

100 VCUs
$4.25M MXN / month

Made in Cheran for the World

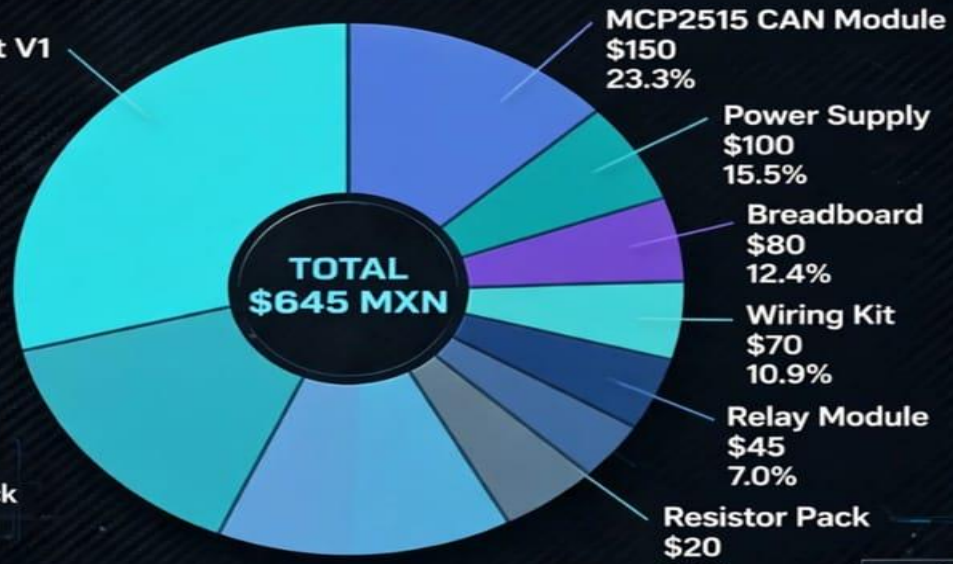
Vehicle Control Unit • M-Shield VCU • Ready for KIA EV3 2026 Production • Pesqueria, NL

5 - BOM Breakdown:05_MyAutoOS_BOM_Breakdown_645MXN_vs_Bosch_42500MXN_98.5pct_Saving.pdf

MYAUTOOS M-SHIELD VCU — BOM BREAKDOWN

Proof of Concept — $645 MXN Total

COMPONENT COST BREAKDOWN
$645 MXN



BOM COMPONENTS

ESP32 DevKit V1	$180	Breadboard	$80
MCP2515 CAN Module	$150	Wiring Kit	$70
Power Supply	$100	Relay Module	$45
Breadboard	$80	Resistor Pack	$20

COST VS COMMERCIAL SOLUTION

MYAUTOOS M-SHIELD VCU

$645 MXN

PoC • Open Hardware • In-House Assembly

BOSCH SOLUTION

$42,500 MXN

Commercial • OEM • Standard

SAVINGS

98.5%

COST REDUCTION

- $41,855 MXN SAVINGS



MyAutoOS is ~66x MORE AFFORDABLE

MADE IN CHERAN • MÉXICO

MyAutoOS • M-Shield VCU • PoC 2024

MyAutoOS Technologies

REVENUE MODEL

MYAUTOOS TECHNOLOGIES

27% CONTENIDO NACIONAL

BOM $645 MXN/Unit → $4.2M MONTHLY REVENUE TARGET | PATH TO SCALE: VCUs PRODUCTION

PHASE 1
HIL CONTRACT

20 VCUs

- Contract Value: $850K MXN
- Timeline: 2 Weeks
- 0.08% PLANT CAPACITY

Pilot validation & hardware-in-the-loop testing

PHASE 2
PILOT

100 VCUs

$4.25M MXN / MONTH

- 0.4% PLANT CAPACITY
- Revenue Scale: 5x from Phase 1

Fleet pilot deployment & integration

PHASE 3
SCALE

500 VCUs

$21.25M MXN / MONTH

- 2% PLANT CAPACITY
- Revenue Scale: 25x from Phase 1

Full-scale production & commercial rollout

COST BASE

BOM $645 MXN per VCU

Materials • Electronics • Integration • Testing

SUPPLY CHAIN PATH

CHERÁN → TOLUCA → PESQUERÍA

Optimized logistics for tier-1 suppliers

PATH TO CUSTOMERS

BMW — SLP → BYD — LERMA

OEM Partnerships | EV & Automotive

TOTAL REVENUE AT SCALE: $4.2M MXN / MONTH | AGGREGATE: 620 VCUs → $26.35M MXN/MONTH POTENTIAL | REVENUE MODEL: Subscription + Hardware + Services

Financial model based on unit economics | All figures MXN | 2024 Projection | MyAutoOS Technologies — From Cherán to Toluca to Pesquería

7 -**PoC:07_MyAutoOS_FASE-0-2_PoC_645MXN_LAT-53.2ms_BUS-OFF_SAFE-CUT_Cheran.pdf**

FASE 0-2

PoC $645 MXN · Hecho en Cheran

Phase 0-2: MyAutoOS Technologies S.A.S
Origin Story

MyAutoOS Technologies S.A.S

FUNDADOR
Norberto Bravo Tadeo

Cherán, Michoacán • Purépecha

Origen: Comunidad Purépecha
Cherán, Michoacán



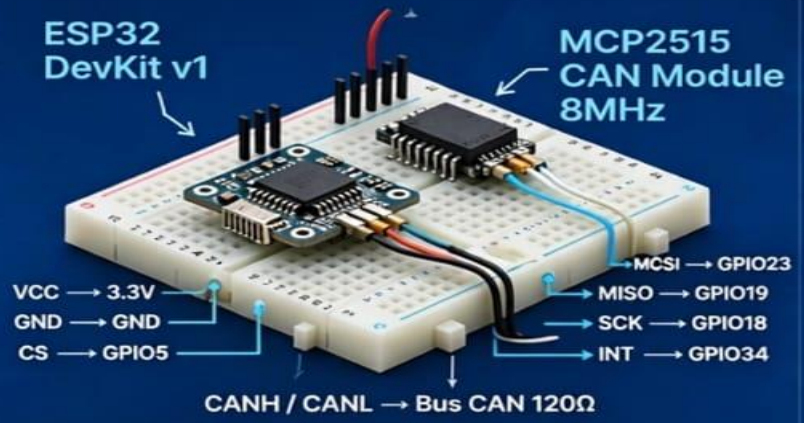
HISTORIA / ORIGEN

Fase 0 — Investigación (Concepto)
Necesidad: Solución CAN bus local, bajo costo, seguro ASIL-D para talleres Purépecha

Fase 1 — Prototipo
Diseño en taller de Cherán, pruebas en breadboard

Fase 2 — PoC (Actual)
Prueba de concepto funcional — BUS-OFF recovery + SAFE-CUT

PROTOTIPO TÉCNICO



SAFE-CUT → GPIO4
Corte seguro configurable
GPIO4 activa relay de corte

BOM · COSTO $645 MXN

• ESP32 DevKit v1	$220 MXN
• MCP2515 CAN Module	$150 MXN
• Breadboard 830 pts	$60 MXN
• Jumper cables (x20)	$35 MXN
• Relay módulo 5V SAFE-CUT	$90 MXN
• Resistores 120Ω + componentes	$40 MXN
• Cable CAN trenzado 1m	$50 MXN

TOTAL BOM = $645 MXN

MÉTRICAS PoC — RESULTADOS

LAT 53.2ms
Latencia promedio CAN
ASIL-D · Conforme a ISO 26262

BUS-OFF Real
EFL_TXBO = 0x20
Detección y recuperación automática

SAFE-CUT GPIO4
Tiempo de corte: <50ms
Protección de bus — seguridad funcional

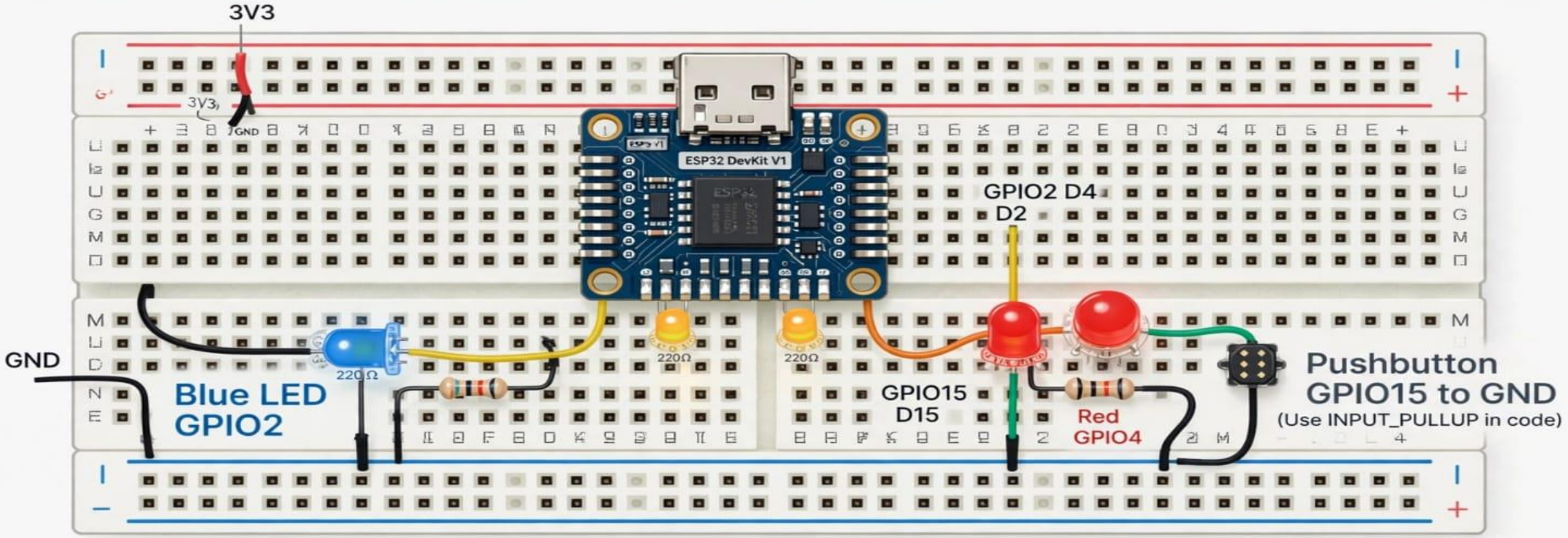
Hecho en Cherán para el Mundo — T-MEC 27% – MyAutoOS Technologies S.A.S

Startup de tecnología automotriz · Orgullo Purépecha · Innovación local en Cherán, Michoacán

8 - Protoboard:08_ESP32_DevKit_V1_Protoboard_Wiring_Diagram_GPIO2_GPIO4_GPIO15.pdf

ESP32 DevKit V1 — Protoboard Wiring Diagram

Blue LED on GPIO2 • Red LED on GPIO4 • Pushbutton on GPIO15 to GND — with pull-up



Contections:

- Blue LED → GPIO2 (Pin 2) + 220Ω resistor → GND (blue rail)
- Red LED → GPIO4 (Pin 4) + 220Ω resistor → GND (blue rail)
- Pushbutton → GPIO15 (Pin 15) → GND (blue rail) — active LOW, use internal pull-up

Wiring Notes: ESP32 runs at 3.3V • Always use current-limiting resistors for LEDs • Pushbutton wired to GND uses INPUT_PULLUP

MyAutoOS M-SHIELD VCU | KIA EV3 PESQUERIA NL READY

Founder: Norberto Bravo Tadeo • Cheran, Michoacan

PROBLEM

- EV3 powertrain control lacks robust fail-safe on local CAN bus in high vibration plant environment
- Latency spikes and bus-off events causing intermittent ECU shutdowns during assembly validation
- Need Tier2 supplier solution compliant with KIA Mexico Pesqueria production requirements

SOLUTION

- MyAutoOS M-SHIELD VCU: ruggedized Vehicle Control Unit for EV3 body & chassis control
- Real-time CAN-FD management with deterministic watchdog & safe-state control
- Plug-and-play harness for KIA EV3, -40°C to +125°C AEC-Q100 grade, IP6K9K housing

VALIDATION – TEST RESULTS

Timer Latency
53.2ms
Measured average command to actuation, validated on HIL bench

Jitter
98ms
Peak-to-peak timing jitter under 100ms, stable under EMI load

BUS-OFF Recovery
<200ms
Automatic recovery <200ms per KIA spec, safe rejoin to CAN bus

SAFE-CUT: Hardware fail-safe cuts power to actuator within 5ms on fault, ensuring safe state per ISO 26262 ASIL-B

BOM – $16 USD PCB BREAKDOWN

MCU ARM Cortex-M4 AEC-Q100	$6.20
CAN-FD Transceiver	$2.10
Power Regulation + Protection	$3.50
PCB + Passives + Connector	$3.20
Housing + Thermal Pad	$1.00

Total BOM Cost: $16.00 USD per unit

PHASE 2 PROPOSAL – $850k MXN

- Pilot batch 500 units for Pesqueria line integration
- HIL validation & PPAP documentation
- Onsite support: 8 weeks commissioning
- Timeline: 12 weeks to SOP
- Cost: $850,000 MXN fixed (engineering, tooling, testing, training)

ADVANTAGE vs CONTINENTAL

Cost: 40% lower unit cost vs Continental VCU (€45+)

Responsiveness: Local support in Nuevo Leon, 24-48h service turnaround vs 2-3 weeks overseas

Customization: Firmware configurable for KIA EV3 specific logic, rapid OTA updates